

- [13] Ruijs CD, Kerkhof AJFM, van der Wal G, Onwuteaka-Philipsen BD. Symptoms, unbearable and the nature of suffering in terminal cancer patients dying at home: a prospective primary care study. *BMC Fam Pract* 2013;14:201.

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Reply to Duffee

Letter to the Editor:

We would like to thank Dr. Duffee for her valuable and thoughtful comments on our work.⁵ She points out 2 important limitations of our approach that must be taken into account when interpreting the results and, even more so, when applying our analytical approach as a proof-of-concept to other research questions.

1. The selection of the text corpus and the distinction between pain-related and general suffering

Dr. Duffee rightly points out that the selection of the text corpus on which our work is based has a decisive influence on the results. She further argues that there is a contradiction between our “concentration on general suffering scholarship” and our goal to define pain-related suffering in particular.

To clarify, our central aim was to examine how suffering was defined in pain research. If an author suggested that his or her theory of (general) suffering also accounted for what it means to suffer from pain, the theory was included, in virtue of being a contribution to pain research. In other words, we remained neutral regarding the relationship of pain-related and general suffering. We are glad, however, that Dr. Duffee highlighted this important point and suggest that she is partly right in calling this problem “philosophical in nature.” A philosophical analysis can help to clarify if suffering from pain is different than suffering from other conditions, and we suggest that the peculiar experiential structure of pain is central to this difference.⁴ However, another way to contrast pain-related and general suffering is to examine the conceptualization of suffering in the respective literature. As we discuss in our article (p. 12), it can be expected that a similar analysis of suffering in a different context will highlight different dimensions and defining aspects.

Thus, we suggest that although some included articles contain theories of general suffering, their presentation in the context of pain research makes them relevant for pain-related suffering.

This being said, we agree that machine learning approaches—while highly useful, if correctly understood as a specific tool—have clear limitations: They can give different definitions and consensus statements with different inputs—which is something the field of artificial intelligence-assisted discourse must grapple with in the future. We suggest that our approach of systematically combining artificial intelligence-based synthesis with conventional qualitative research methods can serve as a use-case for future concept development.

2. The potential influence of historical misconceptions on our analysis and the distinction between suffering and pain

The second point raised by Dr. Duffee concerns a highly charged and essential question in the pain-suffering debate: the precise relationship between suffering, pain, and the underlying neural processes. She rightly emphasizes that

a distinction must be made between nociception, pain, and suffering.

Concretely, Dr. Duffee suggests that we describe an “opposition of suffering to pain by distinguishing the unpleasantness of other pain-related experiences from the aversiveness of pain-related suffering, where the former are tolerable but the latter never is.” She also suggests that this opposition can be traced back to Cassell’s¹ description of pain as “entirely nociceptive.” We will first comment on the author’s historical analysis and then turn towards the problem of intolerability.

(1) Admittedly, we are confused by how Dr. Duffee describes Cassell’s concept of pain in her letter. As she herself points out elsewhere—in a very thorough and instructive publication—Cassell “emphasized that nociception was not pain itself; rather, pain was the interpretation of nociception. As an interpretation, pain was psychological, cognitive, subjective and meaningful.”² However, here, we must remain neutral on what Cassell said or did not say. We happily concede that a definition gained with a procedure like ours can always mirror existing theoretical shortcomings. After all, we developed a methodology to streamline the literature into a representative definition and tried to do so as objectively as possible. We wholeheartedly agree that complementary philosophical and historical discussions are needed and welcome Dr. Duffee’s letter as a helpful contribution to this process.

(2) Dr. Duffee points out that, in the literature, pain and suffering are often contrasted as distinct, even opposed phenomena and criticizes our proposal to use intolerability as characteristic of this difference. She argues that the literature describes “unbearable suffering as a subset of suffering more generally.” To clarify, our analysis does not entail that intolerability is the only difference between suffering and pain. As we discuss, its relation to the self is of vital importance too (p. 12). Furthermore, we would like to suggest that there are differences between different types of intolerability/unbearability that must not be overlooked. We propose using the terms “intolerable” and “unbearable” in a technical sense to point out this difference. The literature Dr. Duffee refers to focuses on palliative situations and some patients’ decision that certain symptoms or situations are *unbearable* in the sense that they prefer to end their lives rather than to bear them. On the other hand, we suggested that suffering will “transform the sufferer into something different from themselves, which is intolerable for them” (p. 12). Suffering can often be lived with, but it is always *intolerable* in the sense that it impinges upon the sufferer a sense that one will be changed in a meaningful and aversive way by whatever one is suffering from. All suffering demands respective action to either stop this change or shape it in a tolerable way, it cannot be tolerated or ignored.

Interestingly, at this point an important parallel to pain emerges. Ecclestone and Crombez offer a “cognitive-affective model of the interruptive function of pain” and describe how pain is preferred “over other demands for attention”³. Pain is adaptive by interrupting and redirecting our focus to stimuli relevant for (physical) survival. Similarly, suffering is adaptive by ensuring a reaction to what we call “perceived threat[s] to an individual’s integrity as a self and identity as a person.” It helps us adapt our sense of self and our personal identity to new situations.

We agree that the debate on the tolerability of pain experiences and the intolerability of suffering needs to be continued. In particular, it is important to better understand the triad of nociception, pain, and suffering and their relationship patterns (Fig. 1).

The Pain-Suffering Complex

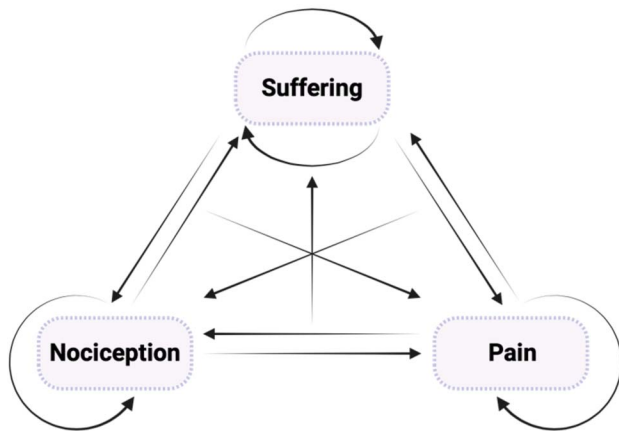


Figure 1. The pain-suffering complex. The triad of pain, nociception, and suffering illustrates the complex relationship between the entities of pain, nociception, and suffering. The arrows between the pillars depict the dynamic and reciprocal interactions and perspectives within this triad. Note that there are 3 distinct phenomenological levels: (1) the phenomenology of suffering itself; (2) the interaction between suffering and nociception, and the interaction between suffering and pain; and (3) the interaction between pain and nociception.

We would like to express our gratitude to Dr. Duffee once again, as her comments will make an important contribution to the future scientific debate on pain-related suffering.

Conflict of interest statement

The authors have no conflicts of interest to declare.

References

- [1] Cassell EJ. Pain and suffering. In: Reich WT, ed. *Encyclopedia of Bioethics*. New York: Macmillan Reference USA, 1995. p. 1897–904.
- [2] Duffee C. An intellectual history of suffering in the *Encyclopedia of Bioethics*, 1978–2014. *Med Humanit* 2021;47:274–82.
- [3] Eccleston C, Crombez G. Pain demands attention: a cognitive-affective model of the interruptive function of pain. *Psychol Bull* 1999;125:356–66.
- [4] Noe-Steinmüller N. Suffering as precarious presence—a husserlian proposal. In: *InterCultural Philosophy*, 2024. doi: 10.11588/icp.2024.1.101135.
- [5] Noe-Steinmüller N, Scherbakov D, Zhuravlyova A, Wager TD, Goldstein P, Tesarz J. Defining suffering in pain: a systematic review on pain-related suffering using natural language processing. *PAIN* 2024;165:1434–49.

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The effect of top-down attention on secondary mechanical hyperalgesia: improvements and controversies?

Letter to the Editor:

We have read with interest the article by Della Porta et al.,¹ recently published in *PAIN*. We are pleased that the authors replicate our findings, showing that a high-load task does not affect the development of secondary mechanical hyperalgesia.^{6,7} According to the authors, the novelty of their study was that they found potential for improvement compared with [our] previous studies and need to address the ongoing controversy. However, our most recent preregistered studies, using large samples,^{6,7} a control condition,⁷ and Bayesian analysis,^{6,7} show no support for the possibility that engaging in a high-load task during sensitization reduces the development of secondary mechanical hyperalgesia at the group level, rather strong Bayesian evidence against it. Hence, the previous literature does not align with the narrative of an ongoing controversy. The authors also incorrectly describe the findings of our first article on this topic,⁸ mentioning that we found a decrease in hyperalgesia in the high-load group when we showed a lack of statistically significant increase in hyperalgesia at the group level (see Figure 3 in Ref. 8 and the discussion).

Della Porta et al.¹ suggest their study has the following improvements: (1) the time-locked presentation of the electrocutaneous stimuli inducing hyperalgesia and the visual stimuli and (2) the use of a different statistical approach.

1. Time-locking

The authors introduce the time-locking aspect as an improvement (or “crucial element” as the authors phrase it), but this is a hypothesis, not a fact. To date, no study has compared the behavioural outcomes of a time-locked vs non-time-locked presentation of the task with electrical conditioning stimuli. The cited studies^{3,4} used a different timing of the visual-nociceptive presentation, a different stimulus (brief lasers stimulus), and measured different outcomes.^{3,4} Furthermore, in 6, we used low-frequency electrocutaneous stimulation, a model in which the electrocutaneous conditioning stimuli are continuously delivered (at 2 Hz), resulting in concomitant electrocutaneous and task administration. Yet, we did not observe any significant effect of the 2-N back difficult task.

The participant’s performance on the difficult task during the administration of the electrical conditioning stimuli was in 1 higher than in 6, but lower than in 7, and similar to 8. Therefore, performance accuracy seems irrelevant for the modulation of hyperalgesia and might also not be a good indication of the engagement in a task. For this reason, in our previous studies, we have complemented behavioral measures with neural and psychophysiological ones.^{6,7} Given that all tasks ended up not modulating the development of hyperalgesia, the statement that time locking is a crucial element is, in this context, not supported by the results. Finally, the ecological validity of the time-locking strategy may be debated, as in daily life, peripheral nociceptive input is not precisely time-locked to the onset of a cognitive task.

2. Statistical test

The authors argue that a 2-way analysis of variance (ANOVA) is an improvement over the “conventional” three-way ANOVA because